

Question

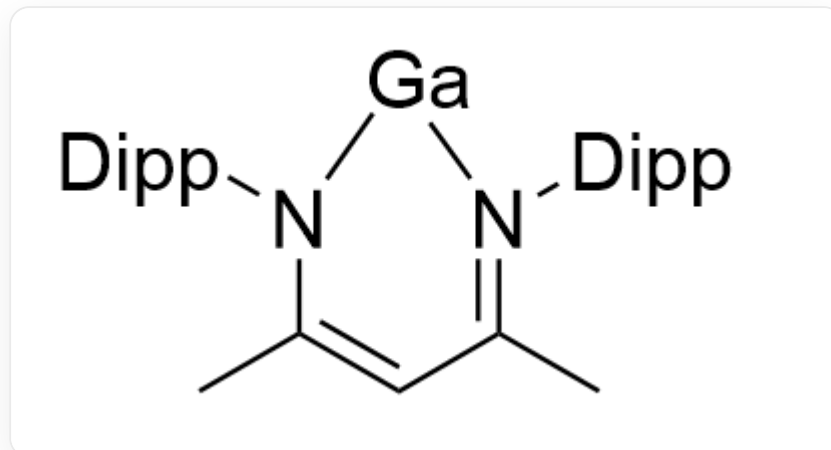
There is a compound **Y** called **Y**:

The cation of **Y** is Bi^{3+} , and **Y** contains only four elements and one anion. The mass fraction of Bi in **Y** is 27.57%, and the mass fraction of C in **Y** is 28.49%.

Y can react with the Ga compound **X** shown in the figure to obtain a compound containing a $-\text{Bi} = \text{Bi}-$ bonded structure, and another product containing the anion of **Y**.

Hint:

1. Generally, the ligands in compounds containing $\text{Bi} = \text{Bi}$ double bonds are not electrically neutral.
2. The 2,6-diisopropylphenyl (i.e., *Dipp*) group connected to the nitrogen atom in **X** has a very large steric volume.



CC1=[N](C2=C(C(C)C)C=CC=C2C(C)C)[Ga]N(C3=C(C(C)C)C=CC=C3C(C)C)C(C)=C1

Which of the following statements are correct:

1. **Y** contains 9 chlorine atoms
2. All *Ga* in the products are 4-coordinated
3. The compound containing $\text{Bi} = \text{Bi}$ double bonds contains $\text{Ga} - \text{O}$ bonds

4. The byproduct without *Bi* contains 40 carbon atoms

5. The stoichiometric ratio of **Y** to **X** is 1 : 1

A. All other options are incorrect

B. 1.2.3.

C. 1.3.

D. 2.3.

E. 1.3.4.

F. 3.4.

G. 3.5.

H. 1.5.

I. 2.5.

J. 1.2.4.

Answer

Correct Answer: **D**

Detailed Explanation

According to the question, assuming **Y** contains one Bi, the number of carbons it contains is:

$$M_Y = \frac{209}{0.2757} \approx 758 \text{ g/mol}$$

$$m_C = 758 \times 0.2849 \approx 216 \text{ g/mol}$$

$$n_C = \frac{216}{12} = 18$$

CHECKPOINT

1 PTS

Assuming **Y** contains one Bi, it contains 18 carbons

The remaining molecular weight is: $758 - 18 \times 12 - 209 = 333$

Consider the element to be *Bi*, which may have three identical ligands, each ligand having 6 *C*s, and a remaining molecular weight of 111, which may be 5 *F*s and one *O*.

CHECKPOINT

1 PTS

The remaining each ligand has 5 F s and one O

Therefore, $\mathbf{Y}$ is $Bi(OC_6F_5)_3$

CHECKPOINT

2 PTS

$\mathbf{Y}$ is $Bi(OC_6F_5)_3$

Therefore, 1 is wrong.

Consider the change in valence: According to the bond structure, Bi is in the +1 valence state and is reduced; in the substrate $\mathbf{X}$, Ga is in the +1 valence state, so it should act as a reducing agent and is generally oxidized to the stable oxidation state $Ga(III)$. To balance the compound's charge, considering that the product contains the anion of $\mathbf{Y}$, the oxidation product should be $GaL(OC_6F_5)_2$, containing 41 carbon atoms, so statement 4 is wrong.

CHECKPOINT

1 PTS

Bi is in the +1 valence state and is reduced

CHECKPOINT

1 PTS

Ga is in the +1 valence state, so it should act as a reducing agent and is generally oxidized to the stable oxidation state $Ga(III)$

CHECKPOINT

1 PTS

The oxidation product should be $GaL(OC_6F_5)_2$

Note that the premise "*Dipp*" is a structure with large steric hindrance. If the organic ligand L of Ga in **X** is directly coordinated with the $Bi = Bi$ double bond to form $L - Bi = Bi - L$, then the spatial distance between the two ligands L is too short, so it cannot be formed due to spatial hindrance. Therefore, the ligand L can only continue to coordinate with Ga ; therefore, the $Bi = Bi$ double bond is likely to be connected to Ga . Therefore, the structure of the reduction product is $L(OC_6F_5)Ga - Bi = Bi - Ga(OC_6F_5)L$. This structure contains an $O - Ga$ bond, so statement 3 is correct.

CHECKPOINT

1 PTS

The spatial distance between the two ligands L of $L - Bi = Bi - L$ is too short, so it cannot be formed due to spatial hindrance

CHECKPOINT

1 PTS

The ligand L can only continue to coordinate with Ga ; the $Bi = Bi$ double bond is likely to be connected to Ga

CHECKPOINT

1 PTS

The structure of the reduction product is $L(OC_6F_5)Ga - Bi = Bi - Ga(OC_6F_5)L$

Therefore, the two product structures that can be obtained are:

```
CC1=[N](C2=C(C(C)C)C=CC=C2C(C)C)[Ga](O[C6F5])(OC#CC#CC#C(F)(F)(F)
(F)F)N(C3=C(C(C)C)C=CC=C3C(C)C)C(C)=C1
```

CHECKPOINT

1 PTS

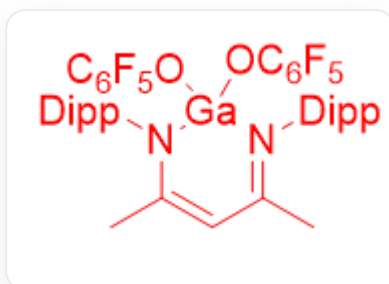
```
CC1=[N](C2=C(C(C)C)C=CC=C2C(C)C)[Ga](O[C6F5])(OC#CC#CC#C(F)(F)(F)
(F)F)N(C3=C(C(C)C)C=CC=C3C(C)C)C(C)=C1
```

```
CC1=[N](C2=C(C(C)C)C=CC=C2C(C)C)[Ga](O[C6F5])([Bi]=[Bi]
[Ga]3(O[C6F5])N(C4=C(C(C)C)C=CC=C4C(C)C)C(C)=CC(C)=
[N]3C5=C(C(C)C)C=CC=C5C(C)C)N(C6=C(C(C)C)C=CC=C6C(C)C)C(C)=C1
```

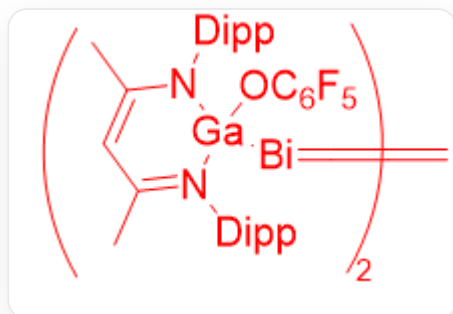
CHECKPOINT

1 PTS

```
CC1=[N](C2=C(C(C)C)C=CC=C2C(C)C)[Ga](O[C6F5])([Bi]=[Bi]
[Ga]3(O[C6F5])N(C4=C(C(C)C)C=CC=C4C(C)C)C(C)=CC(C)=
[N]3C5=C(C(C)C)C=CC=C5C(C)C)N(C6=C(C(C)C)C=CC=C6C(C)C)C(C)=C1
```



```
CC1=[N](C2=C(C(C)C)C=CC=C2C(C)C)[Ga](O[C6F5])(OC#CC#CC#C(F)(F)(F)
(F)F)N(C3=C(C(C)C)C=CC=C3C(C)C)C(C)=C1
```



```
CC1=[N](C2=C(C(C)C)C=CC=C2C(C)C)[Ga](O[C6F5])([Bi]=[Bi]
[Ga]3(O[C6F5])N(C4=C(C(C)C)C=CC=C4C(C)C)C(C)=CC(C)=
[N]3C5=C(C(C)C)C=CC=C5C(C)C)N(C6=C(C(C)C)C=CC=C6C(C)C)C(C)=C1
```

All products are 4-coordinate *Ga*, so statement 2 is correct; both products contain *Ga*, so the reaction stoichiometric ratio should be 2:3, so statement 5 is wrong.

CHECKPOINT

1 PTS

The stoichiometric ratio of **Y** to **X** is 2 : 3

Therefore, 2 and 3 are correct, choose D.